Assignment 4: Relational Database vs. NoSQL Database

COLLEGE MANAGEMENT SYSTEM

DBMS Assignment 4

Database: CollegeManagementSystem

SQL Platform: MySQL 8.0.16 or later

Project scope: departments, faculty, students, courses, and enrollments.

Note: Query outputs described in this report are expected results from the included fictional sample data. Run the SQL script in MySQL Workbench and capture actual screenshots if your instructor requires them.

Introduction

Relational databases store data in tables with defined relationships. NoSQL databases include document, key-value, wide-column, and graph models and often allow more flexible data structures.

Comparison

Aspect	Relational database	NoSQL database
Data model	Tables and relationships	Document, key-value, wide-column, graph
Schema	Usually predefined	Often flexible; varies by product
Transactions	Mature ACID support	Varies; some provide ACID
Scaling	Vertical plus distributed options	Often horizontal scale-out
Use cases	Transactions, joins, structured records	Flexible data, distributed high-volume access

Scalability

Relational systems commonly scale vertically and can scale horizontally through replication, partitioning, or distributed SQL. Many NoSQL systems are designed for horizontal scale-out across multiple nodes.

Consistency

Relational systems commonly provide strong ACID transactions. NoSQL consistency depends on the product and configuration; some provide strong consistency, while others offer tunable or eventual consistency.

Performance

Relational databases are effective for joins, structured queries, and transactional workloads. NoSQL databases can be effective for high-volume key lookups, flexible documents, event data, and distributed workloads when data is modeled to match access patterns.

Examples and use cases

College admissions, course registration, and fee payments suit relational databases such as MySQL or PostgreSQL because of constraints and transactions. A variable product catalogue may suit MongoDB. A session cache may use Redis. A social network graph may use Neo4j.

Why organizations choose different technologies

Selection depends on data shape, transaction requirements, query patterns, latency, throughput, scaling, cost, skills, and compliance. Some organizations use polyglot persistence, selecting different databases for different workloads.

Recommendation

Use MySQL for this College Management System because students, departments, courses, faculty, and enrollments have clear relationships and need referential integrity.

End of Assignment Report